

Idiosyncratic Income Risk and Aggregate Fluctuations

Davide Debortoli, Jordi Gali

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The role idiosyncratic risk plays on aggregate fluctuations depends on how the aggregate fluctuations alter a *consumption weighted average* of *consumption* uncertainty,

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Main Finding

The role idiosyncratic risk plays on aggregate fluctuations depends on how the aggregate fluctuations alter a *consumption weighted average* of *consumption* uncertainty, regardless of whether volatility of idiosyncratic risk varies or not.

- Continuum of households with CRRA (σ) utility.
- Single riskless asset delivering real rate $R_t = \exp\{r_t\}$.
- Idiosyncratic and Aggregate Income Risk.
- Households are only subject to the natural debt limit.

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$$1 = \beta R_t \mathbb{E}_t\{(C_{t+1}(j)/C_t(j))^{-\sigma}\}$$

Up to *second-order* EE can be written as Appendix B.1

$$\mathbb{E}_t \left\{ \frac{\Delta C_{t+1}(j)}{C_t(j)} \right\} \simeq \frac{1}{\sigma} \left(1 - \frac{1}{\beta R_t} \right) + \frac{\sigma + 1}{2} \nu_t(j) \quad (1)$$

where

$$\nu_t(j) := \mathbb{E}_t \{ (\Delta C_{t+1}(j) / C_t(j))^2 \} \simeq \text{var}_t \{ c_{t+1}(j) \}$$

and

$$c_t(j) := \log C_t(j)$$

Appendix B.2 - Exact Version

Appendix C - CARA Formulation

- $\mathcal{O}(|\varepsilon|) :=$ The order of variation in aggregate consumption growth.
- Allow variations in $\nu_t(j)$ to be of order $\mathcal{O}(|\varepsilon|)$
 ... contrasts the representative household case (where $\nu_t \sim \mathcal{O}(|\varepsilon|^2)$)
- Henceforth $\simeq$ represents equalities up to an error of order $\mathcal{O}(|\varepsilon|^2)$

Let $C_t := \int C_t(j) dj$. We are after $c_t := \log C_t$

Aggregating EE:

$$\mathbb{E}_t \left\{ \frac{\Delta C_{t+1}}{C_t} \right\} = \frac{1}{\sigma} \left(1 - \frac{1}{\beta R_t} \right) + \frac{\sigma + 1}{2} \nu_t \quad (2)$$

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$$\underbrace{\nu_t := \int \frac{C_t(j)}{C_t} \nu_t(j) dj}_{\text{Uncertainty Shifter}}$$

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The first-order Taylor expansion of (2):

$$\begin{aligned} c_t &= \mathbb{E}_t \{ C_{t+1} \} - \frac{1}{\sigma} \frac{1}{\beta R} \left(\frac{R_t - R}{R} \right) - \frac{\sigma + 1}{2} (\nu_t - \nu) \\ &= \mathbb{E}_t \{ C_{t+1} \} - \frac{1}{\sigma} \hat{r}_t - \frac{\sigma + 1}{2} \hat{\nu}_t \end{aligned} \quad (3)$$

$$\nu_t := \int \frac{C_t(j)}{C_t} \nu_t(j) dj = \underbrace{\int \nu_t(j) dj}_{\text{Average Effect}} + \underbrace{\text{cov}_j \left\{ \frac{C_t(j)}{C_t}, \nu_t(j) \right\}}_{\text{Comovement Effect}}$$

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Dynamic response of ν_t to a generic aggregate shock ε_t : [Appendix F](#)

$$\frac{d\nu_{t+k}}{d\varepsilon_t} \simeq \frac{d\bar{\nu}_{t+k}}{d\varepsilon_t} + \text{cov}_j \left\{ c_{t+k}(j), \frac{d\nu_{t+k}(j)}{d\varepsilon_t} \right\} \quad (4)$$

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Couple of lessons from this:

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- 1 Idiosyncratic uncertainty has an impact on aggregate consumption only if aggregate shocks have an effect on individual consumption uncertainty.
- 2 The size of the response in ν_t depends on the cross-sectional covariance between consumption and individual uncertainty.

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- ① Idiosyncratic uncertainty has an impact on aggregate consumption only if aggregate shocks have an effect on individual consumption uncertainty.
- ② The size of the response in ν_t depends on the cross-sectional covariance between consumption and individual uncertainty.

$\implies$ Sources of these uncertainty changes are of interest.

Assume the existence of a consumption function

$$c_t(j) = \mathcal{C}(s_t(j), S_t)$$

Let $(\zeta_t(j), \varepsilon_t)$ be i.i.d idiosyncratic and aggregate shocks. Innovation to consumption at t :

$$\begin{aligned}\xi_t(j) &:= c_t(j) - \mathbb{E}_{t-1}\{c_t(j)\} \\ &= f_{t-1}^j(\zeta_t(j), \varepsilon_t)\end{aligned}$$

$f_{t-1}^j(\cdot)$ is some function satisfying $f_{t-1}^j(0, 0) = 0$ and the shocks are scalars.

Individual Consumption Uncertainty: $\nu_t(j) \simeq \mathbb{E}_t\{\xi_{t+1}(j)^2\}$

Variations in individual consumption uncertainty driven by ε_t are assumed to be $\mathcal{O}(|\varepsilon|^2)$

Then

$$\nu_t(j) \simeq \psi_t(j)^2 \sigma_\zeta^2$$

where

$$\psi_t(j) = \left. \frac{\partial f_t^j(\zeta_{t+1}(j), \varepsilon_{t+1})}{\partial \zeta_{t+1}(j)} \right|_{(0,0)}$$

Denote $\Theta_t(j) := \psi_t(j)^2$, referred to as (local) square elasticity.

We have arrived at

$$\nu_t \simeq \sigma_\zeta^2 \int \frac{C_t(j)}{C_t} \Theta_t(j) dj$$

Yielding

$$\frac{d\nu_{t+k}}{d\varepsilon_t} \simeq \sigma_\zeta^2 \left[\frac{d\bar{\Theta}_{t+k}}{d\varepsilon_t} + \text{cov}_j \left\{ c_{t+k}(j), \frac{d\Theta_{t+k}(j)}{d\varepsilon_t} \right\} \right] \quad (5)$$

where

$$\bar{\Theta} := \int \Theta(j) dj$$

$$\frac{d\nu_{t+k}}{d\varepsilon_t} \simeq \sigma_\zeta^2 \left[\frac{d\bar{\Theta}_{t+k}}{d\varepsilon_t} + cov_j \left\{ c_{t+k}(j), \frac{d\Theta_{t+k}(j)}{d\varepsilon_t} \right\} \right]$$

¹See Acharya and Dogra (2020), Bayer et. al. (2019), Ravn and Sterk (2020).

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- Uncertainty shifter fluctuates in response to ε_t regardless¹ of the properties of σ_ζ .

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- The overall impact of the change in $\bar{\Theta}$ in response to ε_t tends to be offset by the effect of its covariance with consumption.

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- The overall impact of the change in $\bar{\Theta}$ in response to ε_t tends to be offset by the effect of its covariance with consumption.
- This is because mean square elasticity is more sensitive to shocks for low-consumption households.

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Continuum of households, identical CRRA preferences face the following budget constraint:

$$\begin{aligned}C_t(j) + B_t(j) &\leq B_{t-1}(j)R_{t-1} + Y_t(j) \\ Y_t(j) &= Y_t \exp\{\zeta_t(j)\} \\ B_t(j) &\geq -Y \exp\{\zeta_1\}/r\end{aligned}$$

Aggregate Shock

$\log Y_t \sim AR(1)$, with $\rho_Y \in [0, 1)$

Idiosyncratic Shock

$\zeta_t(j) \sim$ K-State stationary Markov Process

Independent across households

$$\mathbb{E}[\exp\{\zeta_t(j)\}] = 1$$

Equilibrium: $\int C_t(j) dj = Y_t$ and $\int B_t(j) dj = 0$

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Recall previously:

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This is now

$$\hat{r}_t = -\sigma(1 - \rho_Y)y_t - \sigma\frac{\sigma+1}{2}\hat{v}_t$$

- The sign and size of the impact of an aggregate shock on $\hat{v}_t$ determines the extent to which heterogeneity amplifies/dampens the response of $\hat{r}_t$

Parameter	Meaning	Value
σ	Coefficient of Risk Aversion	1
$\bar{r}$	Steady State Interest Rate (annualized)	0.02
ρ_y	Autocorr. of agg. shocks	0.9
ρ_ζ	Autocorr. of idiosync. earnings	0.966
σ_ζ	Std. Dev. of idiosync. earnings	0.5
n_ζ	Markov Chain Grid Size for ζ	11
n_a	Markov Chain Grid Size for B	500

Table: Calibration of the Endowment Economy

Solution: Endogenous Grid Method + Sequence-Space Jacobian²

²See Auclert et al. (2020)

- SSJ: Finding the first-order approximation of the equilibrium responses to *arbitrary sequences of anticipated shocks* to aggregate endowment over $T < \infty$

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- ★ For the numerical exercise, the exact representation ([Appendix B.2 - Exact Version](#)) is used.

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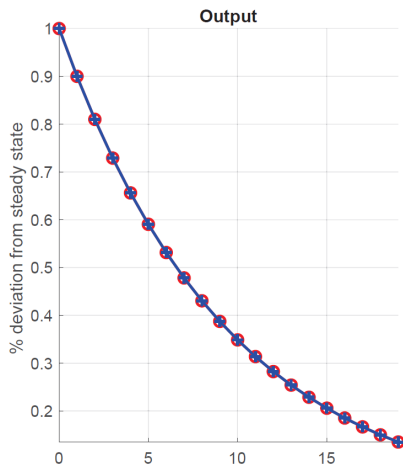
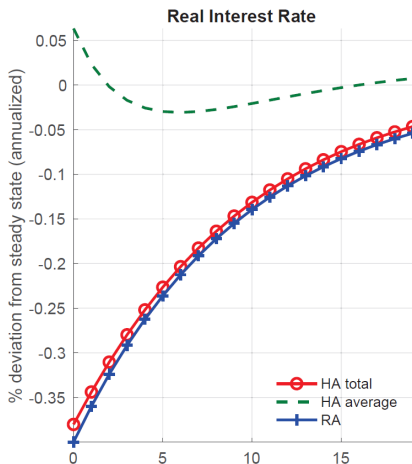


Figure: Endowment Economy - Impulse Responses to 1% Positive Aggregate Endowment Shock

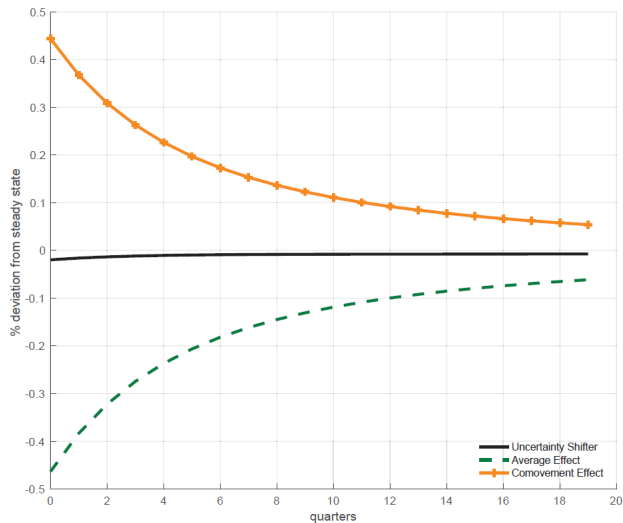


Figure: Endowment Economy - Components of the Uncertainty Shifter

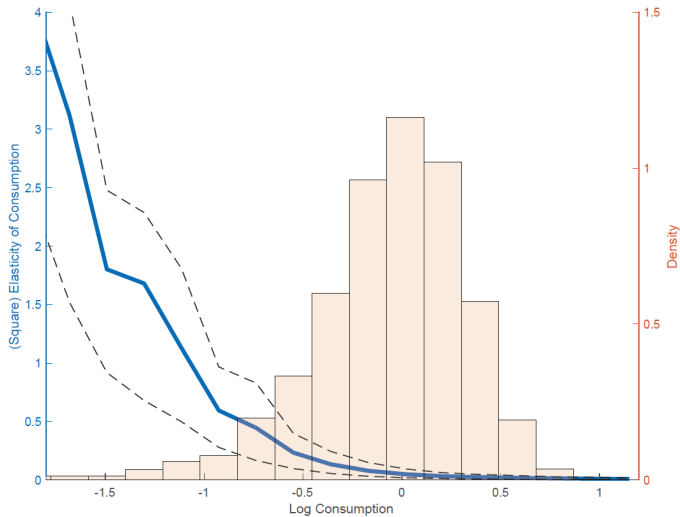


Figure: Endowment Economy - Elasticity of Consumption and Consumption Distribution

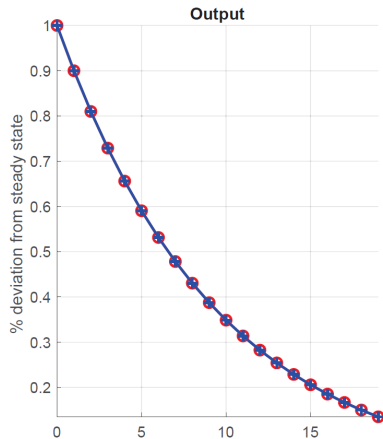
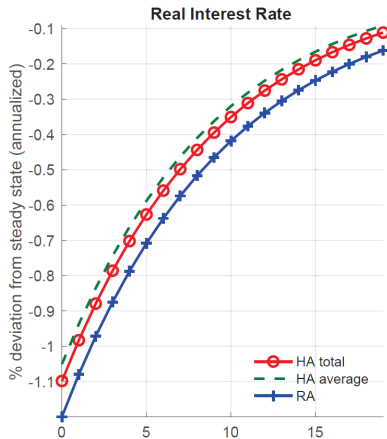


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 $\sigma = 3$

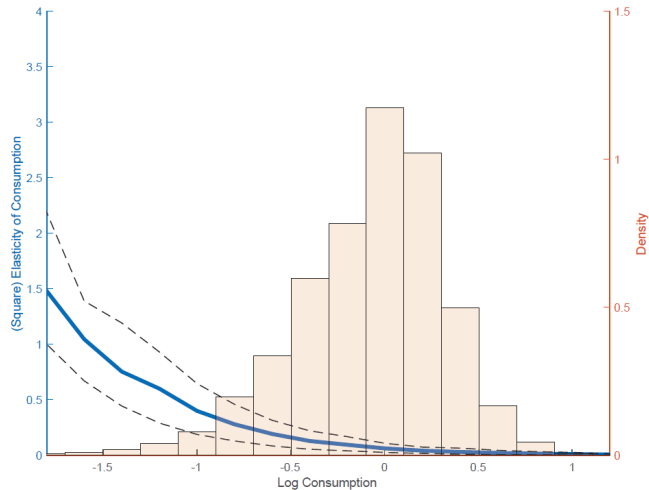


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Several findings that are relevant to the current literature on heterogeneity and aggregate fluctuations:

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- ③ How aggregate shocks affect uncertainty for different types of households is an important empirical question.
... Which is open for exploration!

Continuum of households consume a basket of goods $C_t(j) = (\int C_t(i, j)^{1-1/\varepsilon} di)^{\varepsilon/(\varepsilon-1)}$

Utility $U(C, N) = (\frac{C^{1-\sigma}-1}{1-\sigma} - \frac{N^{1+\psi}}{1+\psi})$

Real wage (per eff. units of labor) W_t and dividends $D_t(j) = D_t$ (non-tradeable)

$$C_t(j) + B_t(j) \leq B_{t-1}(j)R_t + W_t N_t(j) \exp\{\zeta_t(j)\} + D_t(j)$$

Firms: $Y_t(i) = A_t N_t(i)$ where $A_t = \exp\{a_t\}$

Quadratic Adjustment Costs: $\frac{1}{2}\xi P_t Y_t \left(P_t(i)/P_{t-1}(i) - 1 \right)^2$

Labor subsidy τ chosen such that $M^P M^w (1 - \tau) = 1$

Assume CB controls directly the real interest rate: $\hat{r}_t$ follows an exogenous AR(1) process.

Results unaltered if a Taylor-type rule is followed for real interest rate. Appendix D

Up to first-order around zero inflation steady state:

$$y_t = \mathbb{E}_t\{y_{t+1}\} - \frac{1}{\sigma}\hat{r}_t - \frac{\sigma+1}{2}\hat{v}_t$$

$$\implies \hat{y}_t = \frac{-1}{\sigma} \sum_{k=0}^{\infty} \mathbb{E}_t\{\hat{r}_{t+k}\} - \frac{\sigma+1}{2} \sum_{k=0}^{\infty} \mathbb{E}_t\{\hat{v}_{t+k}\} \quad (6)$$

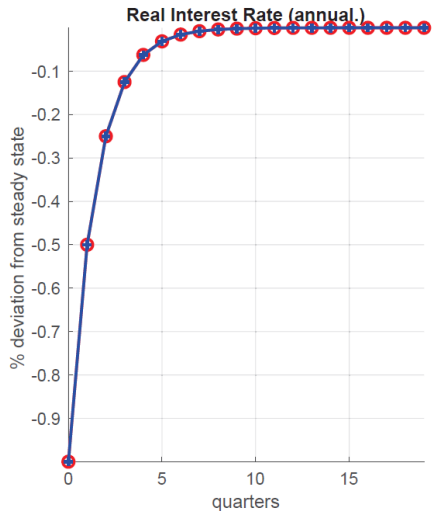
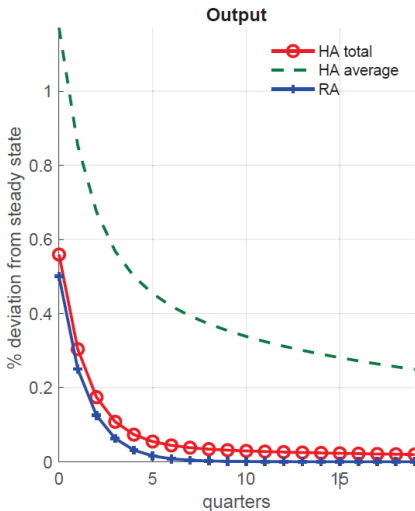


Figure: NK - Impulse Responses to 1% Positive Interest Rate Shock

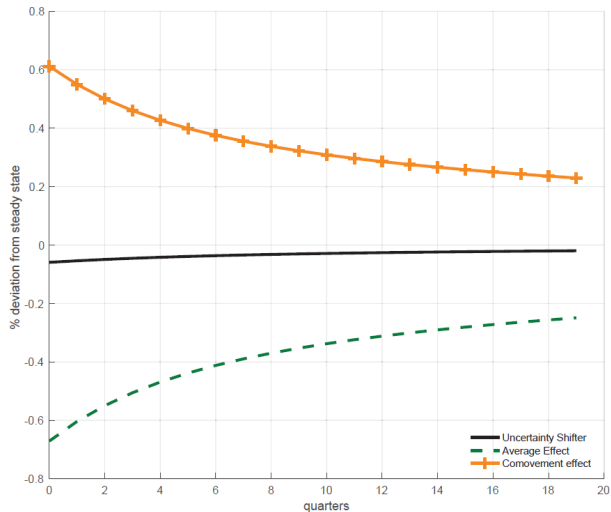


Figure: NK - Components of the Uncertainty Shifter, Interest Rate Shock

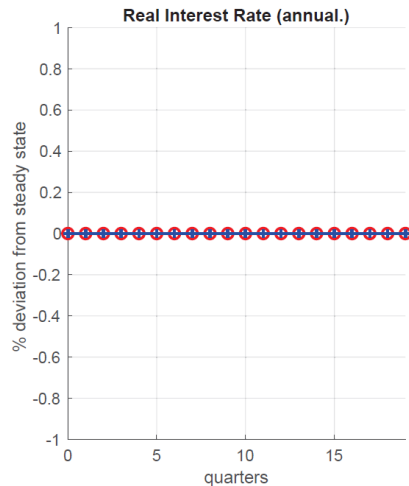
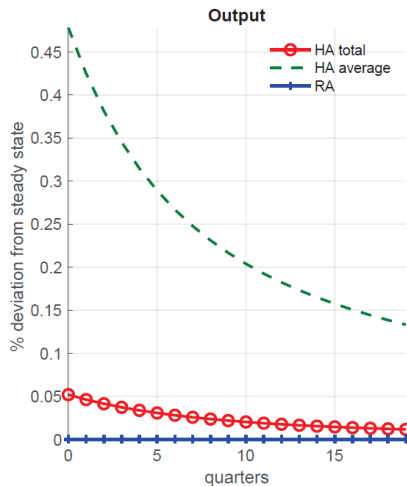


Figure: NK - Impulse Responses to 1% Positive Technology Shock

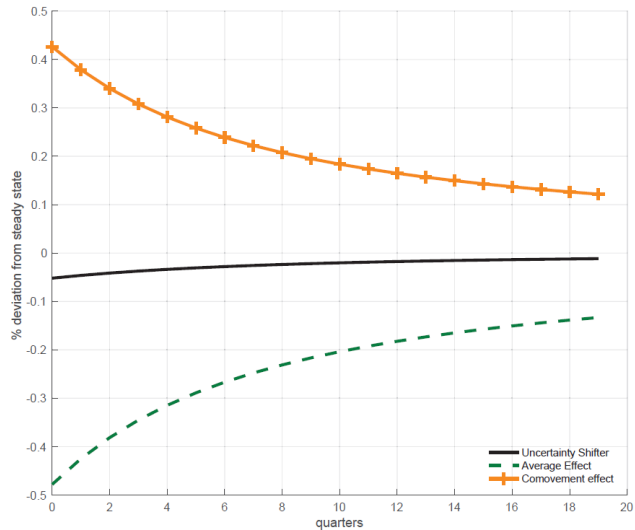


Figure: NK - Components of the Uncertainty Shifter, Productivity Shock

$C_t(j)^{-\sigma} = \beta R_t \mathbb{E}_t \{ C_{t+1}(j)^{-\sigma} \}$, around $C_t(j)$ this is

$$C_t(j)^{-\sigma} \simeq \beta R_t \mathbb{E}_t \left\{ C_t(j)^{-\sigma} - \sigma C_t(j)^{-\sigma} \left(\frac{\Delta C_{t+1}(j)}{C_t(j)} \right) + \frac{\sigma(\sigma+1)}{2} C_t(j)^{-\sigma} \left(\frac{\Delta C_{t+1}(j)}{C_t(j)} \right)^2 \right\}$$

Rearrange:

$$\mathbb{E}_t \left\{ \frac{\Delta C_{t+1}(j)}{C_t(j)} \right\} \simeq \frac{1}{\sigma} \left(1 - \frac{1}{\beta R_t} \right) + \frac{\sigma+1}{2} \nu_t(j)$$

Since $\frac{\Delta C_{t+1}(j)}{C_t(j)} \simeq \Delta c_{t+1}(j) + \frac{1}{2}(\Delta c_{t+1}(j))^2$ write

$$\mathbb{E}_t \{ \Delta c_{t+1}(j) \} \simeq \frac{1}{\sigma} \left(1 - \frac{1}{\beta R_t} \right) + \frac{\sigma}{2} \nu_t(j)$$

First order expanding we get

$$\mathbb{E}_t\{\Delta \mathbf{c}_{t+1}(j)\} \simeq \frac{1}{\sigma} \hat{r}_t + \frac{\sigma}{2} \hat{\nu}_t(j)$$

with $\hat{r}_t := (\beta R)^{-1}(\frac{R_t - R}{R})$ and $\hat{\nu}_t(j) := \nu_t(j) - \mathbb{E}\{\nu_t(j)\}$. We have achieved

$$\mathbb{E}_t\{\Delta \mathbf{c}_{t+1}(j)\}^2 \sim \mathcal{O}(|\varepsilon|^2)$$

bringing

$$\nu_t(j) := \mathbb{E}_t\left\{\left(\frac{\Delta \mathbf{c}_{t+1}(j)}{\mathbf{c}_t(j)}\right)^2\right\} \simeq \text{var}_t\{\mathbf{c}_{t+1}(j)\}$$

One can write

$$C_t(j)^{-\sigma} = \beta R_t \mathbb{E}_t\{C_{t+1}(j)^{-\sigma}\} \iff C_t(j) V_t(j) = (\beta R_t)^{\frac{-1}{\sigma}} \mathbb{E}_t\{C_{t+1}(j)\}$$

where

$$V_t(j) := \left(\frac{\mathbb{E}_t\{C_{t+1}(j)^{-\sigma}\}}{[\mathbb{E}_t\{C_{t+1}(j)\}]^{-\sigma}} \right)^{\frac{1}{\sigma}} \geq 1$$

Then

$$C_t = (\beta R_t)^{-\frac{1}{\sigma}} C_{t+1} V_t^{-1}$$

with $V_t := \int \frac{C_t(j)}{C_t} V_t(j) dj$. In terms of log-deviations from steady state this brings

$$\hat{c}_t = \hat{c}_{t+1} - \frac{1}{\sigma} \hat{r}_t - \hat{v}_t$$

Doing things more generally:

$$\mathbb{E}_t \left\{ \frac{\Delta C_{t+1}(j)}{C_t(j)} \right\} \simeq \frac{1}{\sigma_t(j)} \left(1 - \frac{1}{\beta R_t} \right) + \frac{\gamma_t}{2} \mathbb{E}_t \left\{ \left(\frac{\Delta C_{t+1}(j)}{C_t(j)} \right)^2 \right\}$$

where $\sigma_t :=$ Relative Risk Aversion, $\gamma_t :=$ Relative Prudence. CARA(σ) brings this to

$$\mathbb{E}_t \Delta C_{t+1}(j) \simeq \frac{1}{\sigma} \left(1 - \frac{1}{\beta R_t} \right) - \frac{\sigma}{2} \tilde{\nu}_t(j)$$

with $\tilde{\nu}_t(j) := \mathbb{E}_t \{ \Delta C_{t+1}(j)^2 \}$. Then, if $x_t(j)$ denotes cash-on hand and μ_t the MPC, we have $C_t(j) = C_t + \mu_t x_t(j)$.

But then uncertainty, with iid idiosyncratic income shocks $y_t(j) \sim N(0, \sigma_{y,t})$, $\tilde{\nu}_t(j) = \mu_{t+1}^2 \sigma_{y,t}^2$ is **constant** across households, the comovement effect disappears.

Repeat the NK example with

$$\hat{r}_t = \phi_\pi \pi_t + m_t$$

where m_t is a monetary shock, assumed AR(1) with $\rho_m = 0.5$. Set $\phi_\pi = 0.5$ as in Taylor (1999).

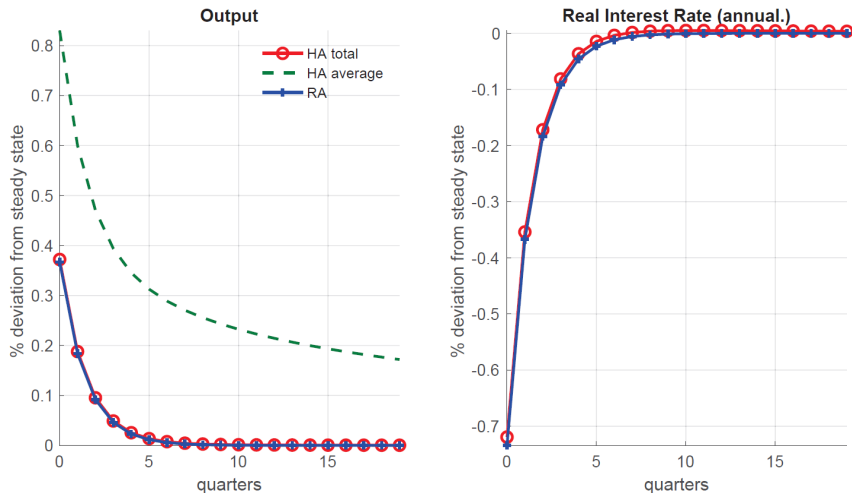


Figure: NK Monetary Rule - Impulse Responses to a 25bp Monetary Shock

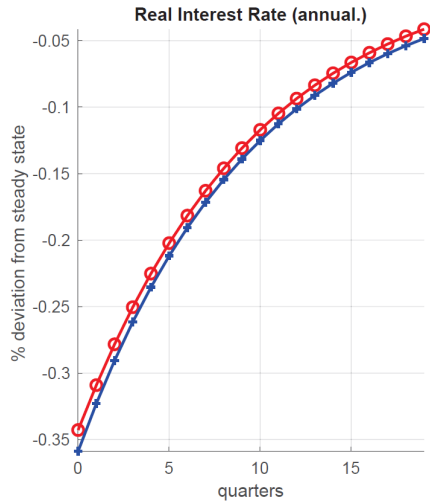
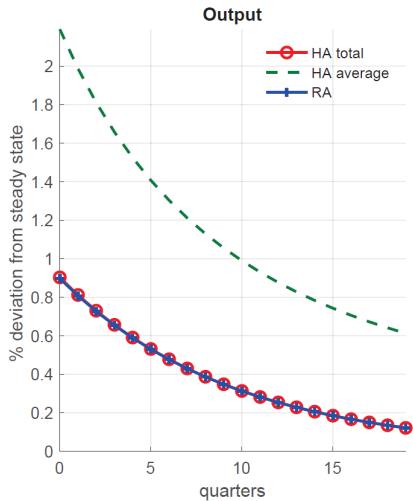


Figure: NK Monetary Rule - Impulse Responses to 1% Positive Technology Shock

Repeat the NK example with the idiosyncratic income process as in Kaplan, Moll and Violante (2018)

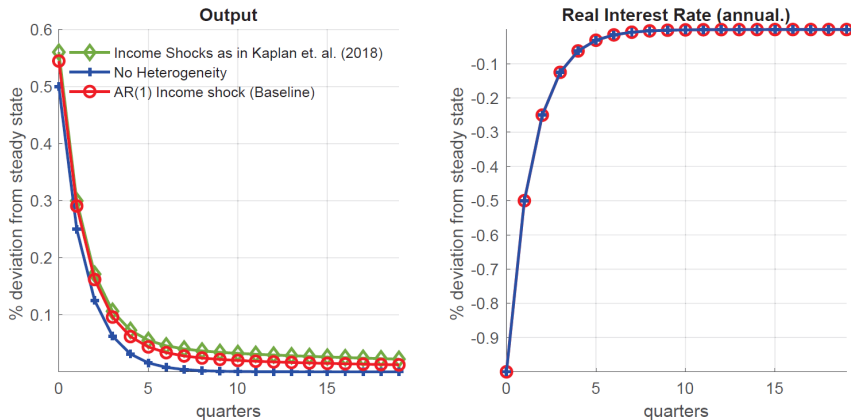


Figure: NK - Income process as in Kaplan, Moll and Violante (2018) - 25bp Monetary Shock

Recall $\nu_t := \int \frac{C_t(j)}{C_t} \nu_t(j) dj$. Thus

$$\frac{d\nu_{t+k}}{d\varepsilon_t} = \int \frac{d[C_{t+k}(j)/C_{t+k}]}{d\varepsilon_t} \nu_t(j) dj + \int \frac{C_{t+k}(j)}{C_{t+k}} \frac{d\nu_{t+k}(j)}{d\varepsilon_t} dj$$

We will argue the first term is $\mathcal{O}(|\varepsilon|^2)$ and ignore it to conclude. Notice we have

$$c_t(j) - c_t = \mathbb{E}_t\{(c_{t+1}(j) - c_{t+1})\} - \frac{\sigma}{2} \nu_t(j) + \frac{\sigma + 1}{2} \nu_t$$

Solve forward:

$$c_t(j) - c_t = - \sum_{k=0}^{\infty} \left[\frac{\sigma}{2} \mathbb{E}_t\{\nu_{t+k}(j)\} + \frac{\sigma + 1}{2} \mathbb{E}_t\{\nu_{t+k}\} \right] + \mathbb{E}\{c_t(j) - c_t\}$$

We can then see

$$\frac{d[c_t(j) - c_t]}{d\varepsilon_t} = - \sum_{k=0}^{\infty} \left[\frac{\sigma}{2} \frac{d\nu_{t+h}(j)}{d\varepsilon_t} + \frac{\sigma + 1}{2} \frac{d\nu_{t+h}}{d\varepsilon_t} \right] \sim \mathcal{O}(|\varepsilon|)$$

Accordingly,

$$\int \frac{d[C_{t+k}(j)/C_{t+k}]}{d\varepsilon_t} \nu_t(j) dj \sim \mathcal{O}(|\varepsilon|^2)$$

Hence,

$$\frac{d\nu_{t+k}}{d\varepsilon_t} = \int \frac{C_{t+k}(j)}{C_{t+k}} \frac{d\nu_{t+k}(j)}{d\varepsilon_t} dj$$